

VFR : Variable Fixed Interest Rate Model

Glovin, Ry, Madiha

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Abstract. This paper introduces the Variable Fixed Rate (VFR) model, an interest rate framework designed to enhance traditional utilization-based interest rate models in decentralized finance (DeFi). The VFR model integrates fixed interest rate positions with variable market influences, providing borrowers with stable rates while allowing lenders to benefit from market fluctuations, thereby balancing the interests of both parties and optimizing capital efficiency.

The VFR model features semi-fungible positions, enhancing liquidity and simplifying position management. It seamlessly integrates with existing pool-based loan models, facilitating market rate discovery and enabling arbitrage opportunities. Additionally, the VFR Swap mechanism allows borrowers to trade interest rate positions through an auction system, with premiums compensating lenders for opportunity costs in high-interest environments.

1 Introduction

Fixed rates present an all or nothing proposition for transferring the risk or opportunity cost towards one party. Especially when the fixed interest rate is set by parameters, this disparity makes it difficult to balance the needs of both parties effectively. Stablecoins present a unique scenario where compensation mechanisms differ. Instead of directly compensating lenders, stablecoins typically reward liquidity providers (LPs) or demand holders within the protocol. This approach separates the demand control mechanism from the direct lending process, as there is no direct counterparty to manage. Instead, holders are often compensated with tokens, which differs from the counterparty dynamic of controlling the lender who provides the actual liquidity for borrowing. Although there were approaches towards tokenizing borrowing rights, borrowing and lending within pool-based systems remain the most intuitive and efficient. Tokenizing borrowing rights requires additional liquidity and interactions with external protocols, which can be counterintuitive and less seamless.

Variable Fixed Rate is an additional layer on top of an existing Pool Based Loan that operates on a utilisation based interest rate model, providing an additional option for borrowing. The user can choose a variable rate that is constantly changing or a fixed rate on top of the VFR that is fixed at the current utilisation based interest rate.

Fixed rate positions on top of the VFR, known as ‘slots’, are traded on VFR Swap, where users who borrow at a variable rate can auction for a fixed interest rate slot at a lower rate with a matching amount. The interest rate swap between the buyer’s variable rate and the fixed rate occurs when the buyer pays a higher amount than the bid paid when the existing fixed rate was purchased, and the existing borrower moves to a variable rate that is more or less in sync with the current market rate, ensuring that the borrower’s interest rate is always comparable to or better than the current market rate. In addition, the premium over the fixed rate paid by the borrower is included in the lending interest and tends to increase as the market rate increases, effectively solving the trade-off of a traditional parameterised fixed rate.

2 Existing Pool-Based Loans

The utilization-based interest rate model, initially introduced by Compound, has undergone significant evolution, most notably with the developments brought by Aave. Their models introduced optimal utilization-based kinks and later evolved into time-based dynamic interest rate curves in which the gradient adjust up and downwards over time, a concept popularized by protocols such as Euler and Fraxlend.

The fundamental principle behind these models is to dynamically adjust interest rates based on the pool’s utilization. When utilization is high, leading to pool illiquidity, the model increases both the borrow interest rate and the deposit interest rate. This dual adjustment discourages new borrowing while encouraging new deposits, thus diluting the pool’s utilization. Conversely, when utilization is low and capital efficiency is suboptimal, the model decreases interest rates to stimulate borrowing and dissuade deposits, thereby increasing utilization and enhancing capital efficiency.

The simplicity of the model ensures intuitive dynamics for both borrowing and depositing activities. As the borrow interest rate increases, the deposit rate rises in tandem, creating an inverse relationship that reduces borrower demand while simultaneously increasing depositor demand. This interplay aligns incentives naturally, promoting balanced utilization of the pool.

Effective risk management in utilization-based interest rate models involves balancing capital efficiency with risk. This balance is primarily achieved through managing the pool’s utilization to avoid illiquidity while ensuring optimal capital use. This process is a critical macro risk parameter for lending pools, as maintaining adequate liquidity is essential for the stability of the protocol. While it is not within the scope of our research, on top of this macro layer, asset-specific risk underwriting adds another dimension to risk management, providing a detailed framework for assessing and mitigating risks associated with individual assets.

One distinctive feature of this model is that borrowers are not charged a fixed interest rate at the time of borrowing. Instead, they are subject to a continuously fluctuating interest rate, determined by the equilibrium between borrowing and lending activities. While this mechanism can effectively adjust the

supply-demand dynamic through interest rate variations, it imposes a continuous monitoring requirement on the borrower. Borrowers need to stay vigilant about rate changes to manage their costs effectively. If users fail to monitor and respond to rate fluctuations, the protocol might struggle to maintain optimal utilization. Another significant aspect of this model's efficiency lies in the fungibility of positions on the utilization-based interest curve. This characteristic enhances liquidity and market efficiency. All positions within the pool are fungible because they receive the same interest rate, which is determined by the pool's utilization rate.

Such an interest rate model can be likened to a bonding curve. In this model, as the demand for fungible positions increases, the interest rate adjusts accordingly. This self-regulating mechanism ensures that the interest rate reflects real-time market conditions, promoting efficient allocation of resources. This design parallels the dynamics of an automated market maker (AMM) or an order book. These systems allow users to trade assets asynchronously, meaning transactions can occur without the need for direct negotiation between counterparties. This asynchronous trading is facilitated by smart contracts that automatically match buy and sell orders. In the utilization based interest rate model, borrowers and lenders interact through a similar mechanism, allowing for instant borrowing and lending without the need to find a counterparty at the time of demand or supply. This reduces friction in the market and increases overall efficiency, as participants can engage with the system at their convenience. Another key factor contributing to the efficiency of this model lies on the perpetual nature of loan positions. Unlike traditional loans with fixed durations and maturities, positions in this model are rolled over every block with an updated interest rate.

3 Existing Fixed Rate Loans

Aave introduced an approach to enhance user flexibility by integrating a nearly stable interest rate model alongside its variable interest rate curve. While the model aimed to provide a 'stable' rate, it only offered competitive rate when the variable rate remained below it. As a result, it didn't achieve a truly 'fixed' or highly competitive rate, which led to limited adoption and its eventual deprecation in Aave V4 [1].

Liquity V2 allows users to set the interest rate [2], and sorts the redemption queue by interest rate, so that when BOLD is de-pegged, redemption happens from the lowest interest rate. Once BOLD starts to de-peg, users will start to bid up the interest rate to avoid redemption, and the increased interest rate paid will be used to incentivize demand to hold BOLD, helping to reverse the peg. The key takeaway here is that the interest rate is discovered and fixed by the user, and the threshold at which the position is closed drives the user to take a specific action in a specific situation - this ensures that the user doesn't set the interest rate too low, but rather drives a situation where the user pays as much as they are willing to pay under current market conditions, thereby allowing

the protocol to not have governance making decisions over interest rates but crowdsourcing it from the users.

RAI [3] and HAI [4] adjusts its redemption rate in response to market prices, decreasing the redemption price when the market price is higher to incentivize borrowing. Furthermore, RAI incentivizes users to mint or repay based on fluctuations in the redemption rate, creating arbitrage opportunities aligned with market conditions. This approach enables arbitrage opportunities based on average market interest rates, promoting efficient market dynamics.

Lending and borrowing on a peer-to-peer basis, exemplified by platforms such as NFTFi [5] and Astar [6], is facilitated through direct agreements between borrowers and lenders. A notable drawback of this approach is that positions are non-fungible and cannot be aggregated. Consequently, lenders and borrowers are unable to lend or borrow asynchronously, limiting their ability to engage in transactions immediately upon their desire.

On the other hand, the approach of tokenizing a borrowing right with a fixed maturity and fixed rate in the form of a coupon bond adds too much complexity to the interest rate model itself, as well as not being intuitive to the user experience, as it requires minting and swapping rights and managing liquidity pools for this model has shown difficulty in maintaining sufficient liquidity.

MakerDAO [7] provides governance-defined interest rates. The DAO aims to update the rate as infrequently as possible, often only making small adjustments over time, and due to the cadence of governance, users are aware when new rates will be introduced and their value before time (2 weeks as the time of writing), this provides “predictable” rates. The rates are still sensitive to market conditions and in the case that the market moves fairly quickly, governance will take emergency actions to modify rates.

4 VFR Interest Rate Design

4.1 Design Principles

1) Fixed Rates with Variable Market Influence Each borrower’s interest rate is determined at the time of opening their position and remains fixed until repayment or forced closure. Although market interest rates fluctuate, the fixed rate for each borrower ensures predictability and stability for the duration of their loan.

2) Balancing the interests of both parties While borrowers enjoy fixed interest rates, lenders benefit from market conditions as their yields adjust in response to general interest rate trends. When market interest rates rise, lenders see increased returns, aligning their incentives with the broader economic environment.

3) Perpetual Loan Positions The VFR model emulates the perpetual loan positions seen in protocols like Aave, where loan positions are continuously rolled

over every block with updated interest rates until a closure condition, such as an unfavorable Loan-to-Value (LTV) ratio, is met. Similarly, in VFR, positions are maintained until the prepaid premium is insufficient to sustain the fixed rate.

4) Fungibility of positions Semi-fungible positions in VFR offer a balance between interchangeability and individualized interest rates. While positions with different interest rates cannot be aggregated due to differing interest rates, they still provide a degree of flexibility. This semi-fungibility enhances liquidity and simplifies position management, leading to more efficient and liquid markets and facilitating effective price discovery.

5) Integration with Utilization-Based Models The VFR system integrates with existing pool-based loan models, adding value without introducing too much unnecessary complexity. This layered approach supports a wide range of borrower preferences and financial scenarios, ensuring that the system remains accessible. It also promotes macro-level market rate discovery and enables arbitrage opportunities based on average market interest rates, paired with fungibility of positions enabling effective balance between capital efficiency and pool illiquidity mitigation with potential to achieve higher utilization of pools when demand is low.

6) Precise Micro-Level Interest Rate Valuation VFR adeptly accommodates user-set interest rates by balancing intrinsic and extrinsic valuations. Intrinsically, it reflects borrowers' personal preferences and risk tolerance. For example, a borrower seeking long-term stability might pay a higher premium for a fixed low interest rate, anticipating future savings. Extrinsically, it incorporates market conditions such as current average rates and future projections. In high-interest environments, the spread between market rates and the borrower's fixed rate highlights the value of rate stability, driving borrower willingness to secure favorable terms.

4.2 Two Modules – VFR and VFR Swap

The Variable Fixed Rate (VFR) model enhances the flexibility of utilization-based interest rate model by adding an additional layer that optimizes the pool's utilization while offering fixed interest rate positions. This model allows borrowers to secure a fixed interest rate at the time of borrowing, which remains constant for the duration of their loan, despite the variable nature of the overall interest rate environment. In essence, while the interest rate for the entire pool fluctuates based on utilization, each individual loan position maintains a fixed rate determined at the moment the loan is taken out. These VFR positions can be tokenized in a semi-fungible manner. For instance, if one borrower holds a \$100 position at a 1% interest rate and another holds a \$50 position at the same

rate, these positions are interchangeable. However, positions with different interest rates are not fungible with each other.

In the VFR swap layer, borrowers can trade their interest rate positions, or slots, through an English Auction, paying a premium to secure these positions. This premium is then transferred to lenders through a smoothing mechanism, which compensates for opportunity costs in high-interest rate environments. As demand for low fixed interest rates increases, this system ensures that lenders receive sufficient returns, even when market conditions drive interest rates higher, thereby maintaining an attractive environment for both borrowers and lenders.

4.3 VFR Design

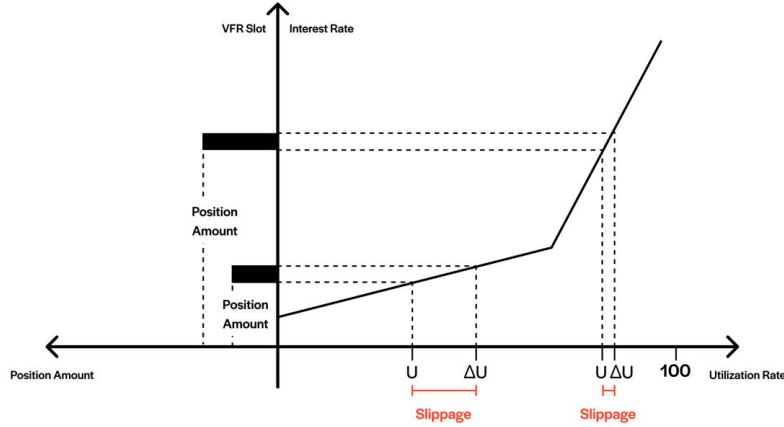


Fig. 1. Overview of Variable Fixed Rate Model

Loan positions opened with VFR will have interest fixed at a current utilization-based interest rate based on a combination of variable rate and VFR plus slippage(see Fig.1).

Here is the interest rate at the time of position opening, taking interest rate slippage into account:

Case 1: $U < U_{Optimal}$

$$InterestRate + Slippage = R_{v0} + \left(\frac{R_{slope1}}{U_{optional}} \right) \times \left(U + \frac{1}{2} \times \frac{BorrowAmount}{TotalSupply} \right)$$

Case 2: $U \geq U_{Optimal}$

$$InterestRate + Slippage = R_{v0} + R_{slope1} + \left(\frac{R_{slope2}}{1 - U_{optional}} \right) \times \left(U - U_{optional} + \frac{1}{2} \times \frac{BorrowAmount}{TotalSupply} \right)$$

If there is a utilization fluctuation, the position will continue to be stacked on top of the corresponding slot. However, since $Interest + Slippage$ is a rational number, it is very rare for these values to match, so we aggregate the amount in intervals r_1 to r_2 based on the corrected interest rate value. The aggregated value $PositionAmount_{r_1 r_2}$ can be defined as follows:

Case 1: $U < U_{Optimal}$

$$PositionAmount_{r_1 r_2} = \int_{r_1}^{r_2} \frac{1}{50} \left\{ (R - R_{v0}) \left(\frac{U_{optional}}{R_{slope1}} \right) - U \right\} \times TotalSupply \cdot dR$$

Case 2: $U \geq U_{Optimal}$

$$PositionAmount_{r_1 r_2} = \int_{r_1}^{r_2} \frac{1}{50} \left\{ (R - R_{v0} - R_{slope1}) \left(\frac{(1 - U_{optional})}{R_{slope2}} \right) - U + U_{optional} \right\} \times TotalSupply \cdot dR$$

Instead of an arbitrary interval, we set the interval at regular intervals from 0, set each interval as a Slot, and set the length of the Slot to determine the SlotSize. If we assume $SlotSize = 10^N$, then the Slot Index is the starting r -value of the slot, which is:

$$SlotIndex = \lfloor (InterestRate + Slippage) \times 10^N \rfloor / 10^N$$

The accumulated Position Amount per slot divided in this way is as follows:

Case 1: $U < U_{Optimal}$

$$PositionAmount_{slot} = \int_{index}^{index+slotSize} \frac{1}{50} \left\{ (R - R_{v0}) \left(\frac{U_{optional}}{R_{slope1}} \right) - U \right\} \times TotalSupply \cdot dR$$

Case 2: $U \geq U_{Optimal}$

$$PositionAmount_{slot} = \int_{index}^{index+slotSize} \frac{1}{50} \left\{ (R - R_{v0} - R_{slope1}) \left(\frac{(1 - U_{optional})}{R_{slope2}} \right) - U + U_{optional} \right\} \times TotalSupply \cdot dR$$

The resulting $PositionAmount$ will be the $BorrowAmount$ Cap for the Slot. However, the interest accumulated will be less than the existing interest by applying vr to the entire $BorrowAmount$ (see Fig.2). This difference is the opportunity cost that needs to be compensated to the Lender:

$$CompensateAmount = R_{market} \times U \times TotalSupply / 100 - \sum PositionAmount_{slot}$$

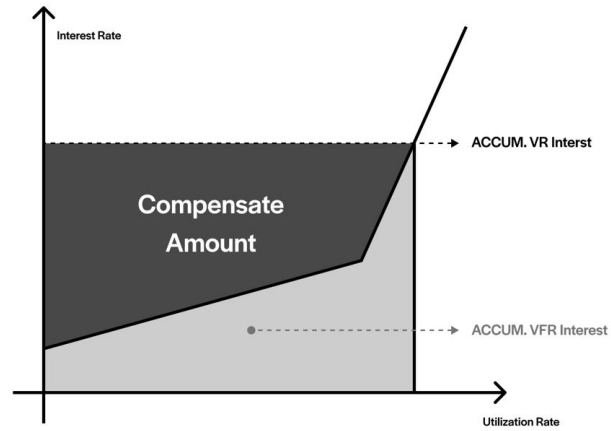


Fig. 2. Opportunity cost of lenders in VFR

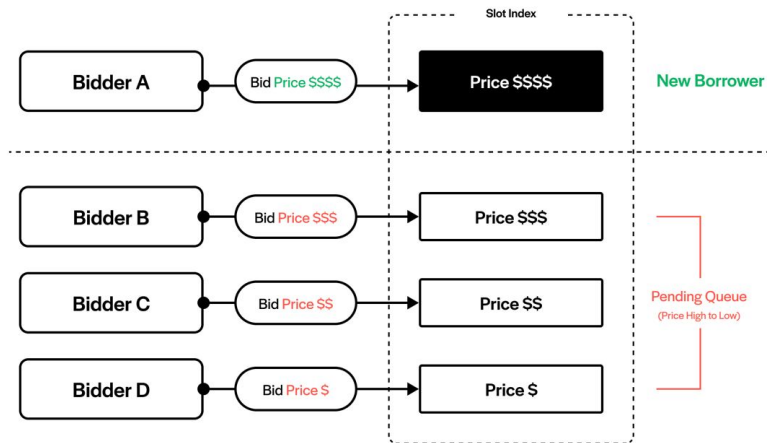


Fig. 3. Bid Queue Structure

4.4 VFR Swap Design

The interest rate fixed by the borrowers through the VFR interest rate model is aggregated and stacked on top of each slot and placed on the corresponding auction queue(see Fig.3) and we also propose two potential auction methods with different trade-offs. In a high interest rate situation, or if there is a slot with an interest rate lower than the current interest rate, the variable rate borrower can submit a bid for each slot, and when a price higher than the current bid is submitted, an interest rate swap will take place between the VFR and VR_{market} fixed on the slot. The *CompensateAmount* will be the amount to be covered through the paid premium between borrowers in slots with lower and higher interest rates.

1) Refund Auction Method Borrower places a bid for the amount of slot he/she is willing to pay. If the bid is higher than the bid of the person who currently owns the slot, he/she wins the slot, and the former owner receives the bid he/she paid back. At this time, the lender is compensated with the difference caused by the renewal of the bid.

Assumption: Given an index of r , the previous winner A 's bid for the slot at the on-time price, $SlotPrice_{rA}$, and the new winner B 's offer, $SlotPrice_{rB}$, the $SlotTradingFee_r$, the amount of compensation paid to the lender when the slot is traded, is:

$$SlotTradingFee_r = SlotPrice_{rA} - SlotPrice_{rB}$$

2) Full-Payment Auction Method Borrowers bid the amount they think the slot is worth. All paid bids will be paid to the lender as a reward, and the person who bids the highest slot wins.

Assumption: If the amount offered by $bidder_i$ out of n bidders for a slot with index r is called $SlotPrice_{ri}$, the $SlotTradingFee_r$, the amount of compensation paid to the lender when the slot is traded, is as follows:

$$SlotTradingFee_r = \sum_{i=1}^n SlotPrice_{ri}$$

Pending Bid Users who wish to bid on a slot but believe that the bid is high place a bid, which is recorded in the Bid Queue from the highest bidder. When the borrower who won the slot closes the position, the top bidder in the Pending Bid Queue of the slot wins the slot and can open the position, and the *PositionAmount* is equal to the *PositionAmount* before closing the position. In this case, the borrower's bid cost incurred by closing the position is not refunded but entirely paid to the lender.

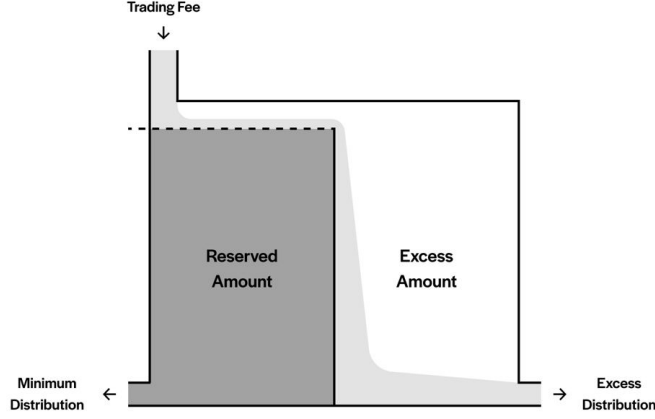


Fig. 4. Smoothing Algorithm

The bid price of a slot will be less than the difference in interest that the user is willing to pay for the difference between the VFR and VR of the slot, so the Slot Price will theoretically converge in the following manner:

$$SlotPrice \approx \max(Principal \times EstimatedBorrowPeriod \times (R_{market} - R_{slot}))$$

Optimally, the Daily Slot Trading Volume should be greater than the Minimum Distribution Amount Per Day value. Thus, lenders are compensated for the opportunity cost of the difference between VFR and VR and receive the excess distribution.

Smoothing Algorithm Paid premiums are distributed according to a smoothing algorithm as distributing them all at once would leave the pool vulnerable to manipulation to snipe this premium. Paid premiums are first accumulated to the Reserved Amount of the Distribution Queue until the Reserve has a value of (Minimum Amount for N days + buffer B%) for each day :

$$MinimumAmountPerDay \times N \times \frac{B}{100}$$

If we set the Minimum Distribution Amount per Day value to Compensate Amount Per Day :

$$MinimumDistributionAmountforDay = \frac{R_{market} \times U \times TotalSupply/100 - \sum PositionAmount_{tick}}{36500}$$

Once there is enough Reserved Amount secured in the queue, the excess is distributed over M days. At a minimum, we distribute the Minimum Amount Per Day + Excess Distribution Amount every day (see Fig.4). Optimally, the Daily Slot Trading Volume should be greater than the Minimum Distribution Amount Per Day value.

5 Discussion

Parameter Ramping The rigidity of locking positions within specific interest rate bands can complicate parameter adjustments. In a model where interest rate stability is crucial, rapid parameter modifications are less desirable. Thus, a gradual interest rate ramping mechanism should be implemented.

Sensitivity of the Model The level of user engagement in interest rate arbitrage and the model's sensitivity to market rate fluctuations are critical factors influencing lender compensation. Attracting more interest rate arbitrageurs would be one way, having a vault with interest rate arbitrage strategies could be another possibility.

Automated Position Managements Users might often lack clarity on the appropriate premium to pay. Therefore, providing tools to calculate the minimum purchase amount for a slot and the maximum savings with an interest rate slot is necessary. An approach that partially automates position management, such as rebalancing via an agent like DeFi Saver, would be beneficial.

VFR as an Additional Layer Leveraging a supplementary approach to an existing model, rather than creating a new one, ensures that users output is always either the same as the market rate or a slightly better, while achieving efficiencies through the fungible positions offered by variable rates. Even if the Variable Fixed Rate is closed due to slot hijacking, users can either maintain or close their positions, as interest rate arbitrage will shift to the Variable Rate, which aligns more closely with the market rate.

Automated Valuation Within the current model, users have significant freedom to express their preferences, driving micro-level interest rate discovery. However, the potential exists to introduce an automated pricing model once market equilibrium is achieved and sufficient data is gathered.

Technical Specifications For better accounting, the VFR model can be

implemented using a tick mechanism to manage discrete interest rate bands, allowing for precise control over fixed interest rates. Each tick maintains its own liquidityIndex to reflect the accrual of interest at that specific rate. When a borrower opens a position, the corresponding tick's liquidityIndex is used to calculate the accrued interest. For instance, suppose we have three ticks representing interest rates of 2%, 3%, and 4%. A borrower opens a position when the utilization rate suggests a 3% interest rate. The system assigns the liquidityIndex that corresponds to the 3% tick. The effective interest rate at position opening should consider the interest rate impact. To account for this impact, the system should also assign the impacted interest rate. For example, if the borrower's position is large enough to affect the interest rate, starting at 3% and ending at 4%, the system should mint DebtTokens corresponding to the 3% interest rate tick for amount A and the 4% interest rate tick for the remaining amount B. In this case, the DebtTokens calculate the total amount due based on the liquidityIndex for each tick.

Enhancing Overall Borrow Rate If paid premium from the purchase of slots in the variable fixed rate model is substantial, it could allow for a broad reduction in the slope of the variable rate without compromising lenders' opportunity costs, thus optimizing interest rate arbitrage opportunities while offering better rates to variable rate users.

Interest Rate Arbitrage In the current high-yield environment, variable fixed rates are particularly advantageous, allowing users to leverage yield-bearing assets as collateral to achieve a positive net APY and lock in costs for a more predictable strategy. Additionally, traditional variable rates require off-chain tracking for interest rate changes for setting certain triggers for changes in net APY. In contrast, the Variable Fixed Rate model allows for immediate action in response to borrowing rate changes due to slot hijacking, enabling a more automated and flexible market response.

Expressing Individuals' Preference This system enables individuals to assign a monetary value to their borrowing position, effectively addressing the question, "How much do you value your borrowing position?" Which can contribute to achieving a higher level of market efficiency.

Directly Derived Fixed Interest Rate From Variable Rate Pool To enhance the accuracy and security of our loan origination process, we have adopted a strategy that leverages the main pool, which operates with a variable interest rate to sync with market rates, rather than independently determining a rate based on utilization at the inception of the loan. This method is designed to mitigate potential exploitation scenarios, such as artificially reducing utilization by depositing a substantial amount of capital into the fixed-rate pool. By subsequently opening a fixed-rate position from a different address, one could otherwise secure a loan at an unduly low rate.

By deriving and anchoring our fixed rates to those of the main variable-rate pool, we maintain alignment with prevailing market conditions and enhance the robustness of our rate-setting mechanism against such manipulative tactics.

Trade-Offs Between Auction Methods The two VFR Swap Auction methods have distinct implications. In a full payment auction, the borrower's financial burden is higher, potentially leading to passive bidding and a prolonged convergence to the highest price. This method may be less time-sensitive but can result in larger payments to lenders. Conversely, in a refund auction, the borrower's financial burden is lower, likely encouraging more active bidding, though this comes at the expense of reduced lender compensation per transaction. The optimal balance between these models should be determined through analysis of market rate sensitivity and borrower participation in bids.

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